Gas Turbine Performance & Predictive Maintenance Analytics

Marine Gas Turbine Engineering Case Study | NASA C-MAPSS FD001

Educational analysis using simulated aircraft turbofan data as a proxy for exploring marine gas turbine condition-monitoring concepts. No actual LM2500 operating data were analyzed.

Executive Summary

This project combines Python, SQLite, machine learning, engineering dashboards, and maintenance-risk assessment to study engine operating-life variability and remaining useful life prediction.

Dataset Overview

• 100 simulated training engines.

• 20,631 operating observations.

• 21 sensor channels and 3 operating settings.

• Average operating life: 206.31 cycles.

• Operating-life range: 128 to 362 cycles.

Predictive Maintenance Results

• Random Forest regression model.

• 80 training engines; 20 held-out validation engines.

• 4,070 validation observations.

• Mean absolute error: 25.71 cycles.

• Root mean squared error: 35.20 cycles.

• R-squared: 0.7126.

• Near-failure RUL overestimation: 75.16%.

Engineering Findings

The simulated fleet exhibited substantial operating-life variation. Selected temperature and pressure sensor channels showed average early-to-late changes across completed histories.

Two independent simulated engines were compared: Engine A operated for 207 cycles and Engine B for 128 cycles. Their 79-cycle difference is descriptive and does not establish a failure cause.

Near-failure predictions frequently overestimated remaining life. This is a significant limitation for potential maintenance decision support.

Engineering Recommendations

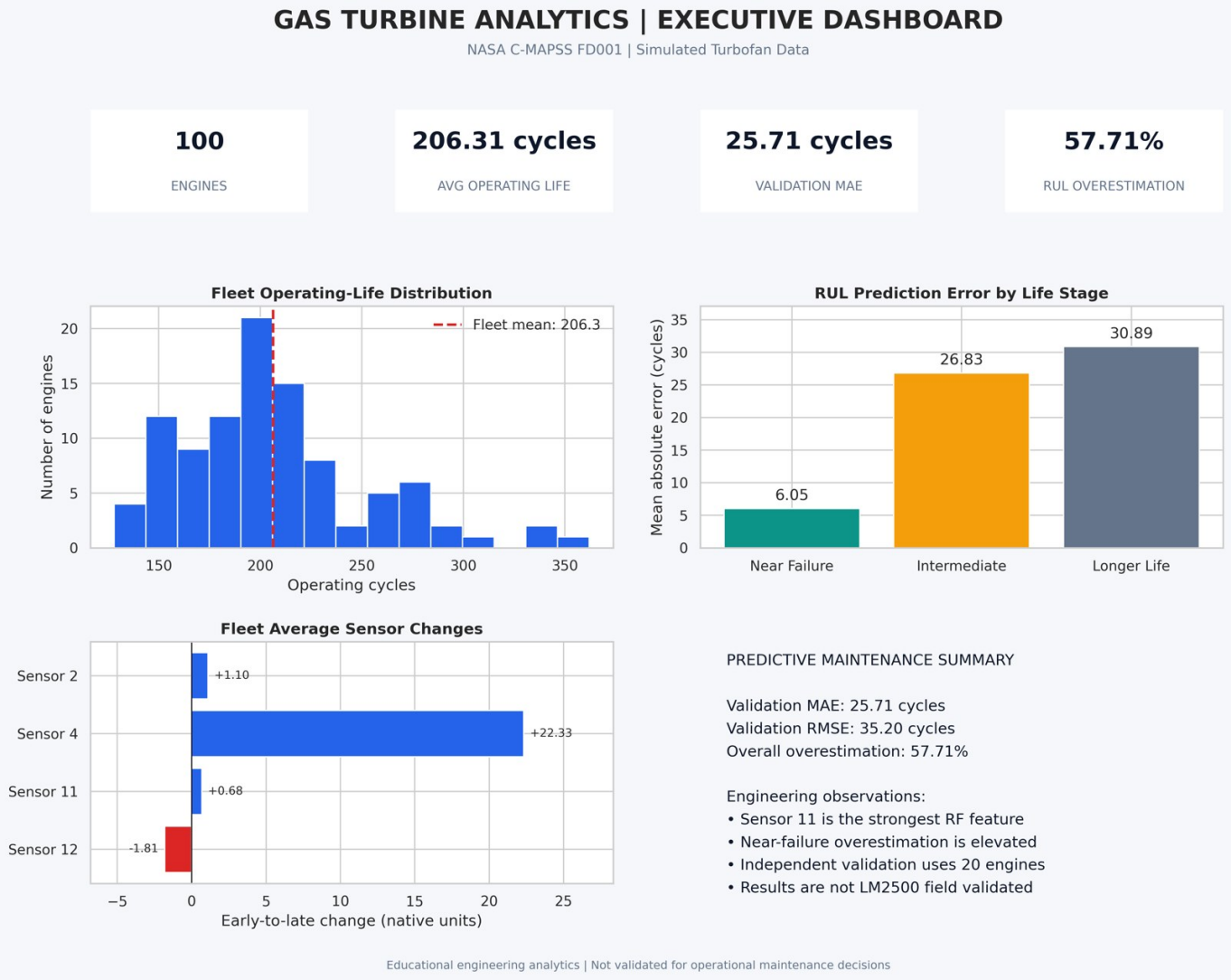
- Investigate sensor trend monitoring.
- Evaluate calibrated RUL uncertainty estimates.
- Compare equipment-level reliability histories.
- Develop an engineer-reviewed inspection queue.
- Improve independent model validation.

Limitations and Safety Boundary

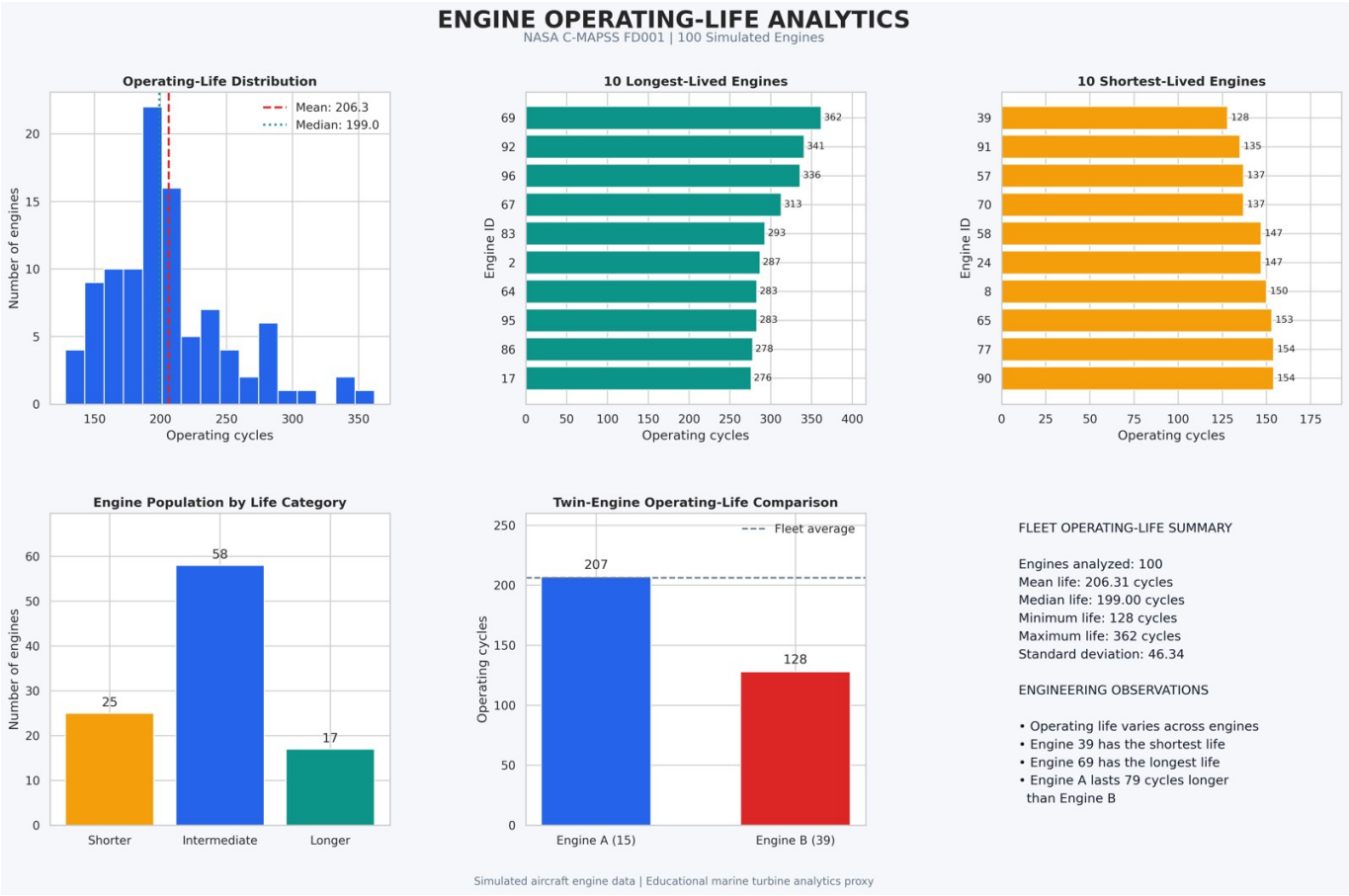
The dataset is simulated aircraft turbofan data, not GE LM2500 operational data. The selected twin-engine predictions are in-sample. No operational alarm thresholds, inspection intervals, or maintenance decisions have been validated.

Actual equipment maintenance must follow approved OEM and site-specific procedures. The proposed analytics framework is advisory and educational.

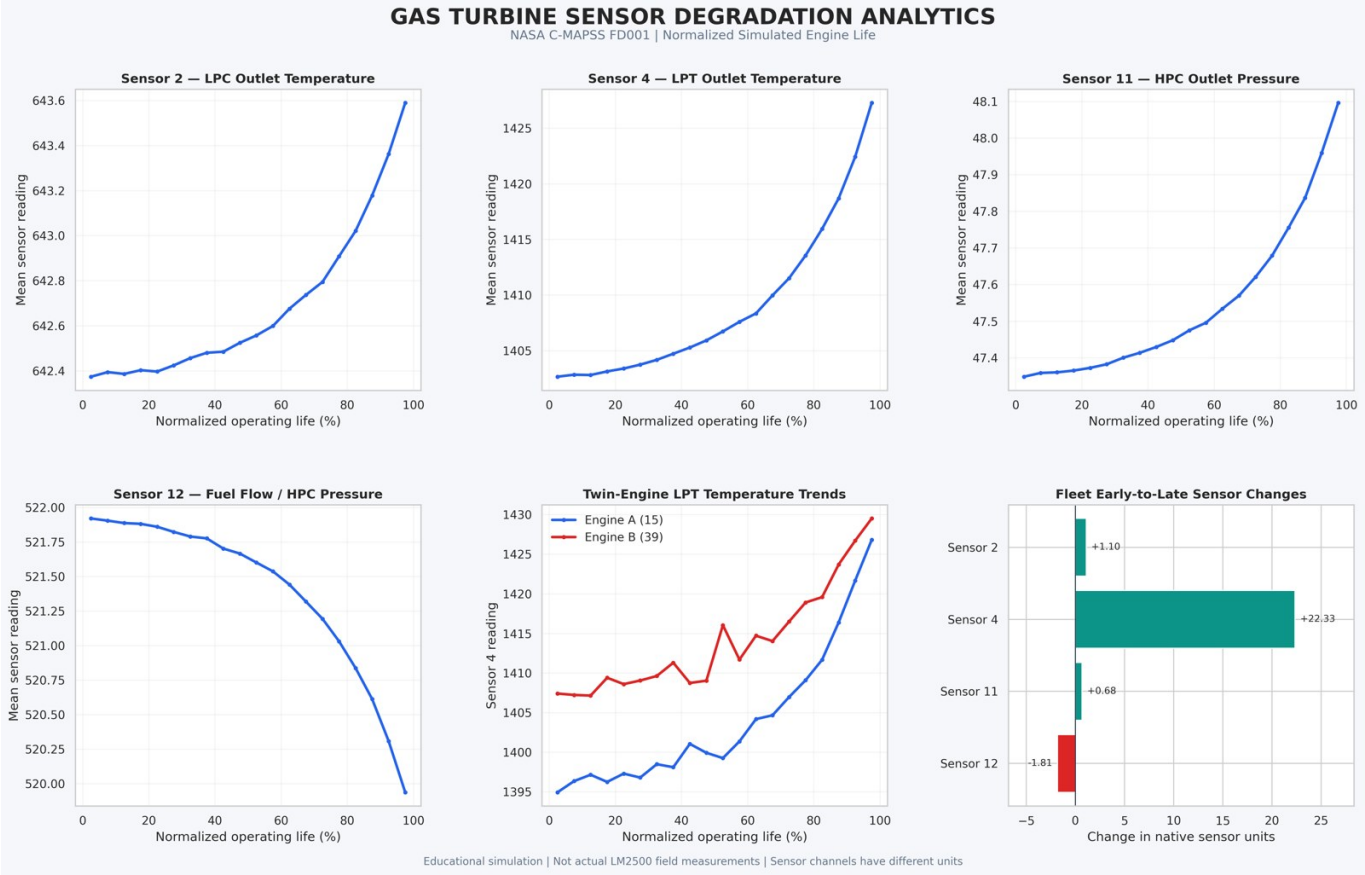
Executive Fleet Overview



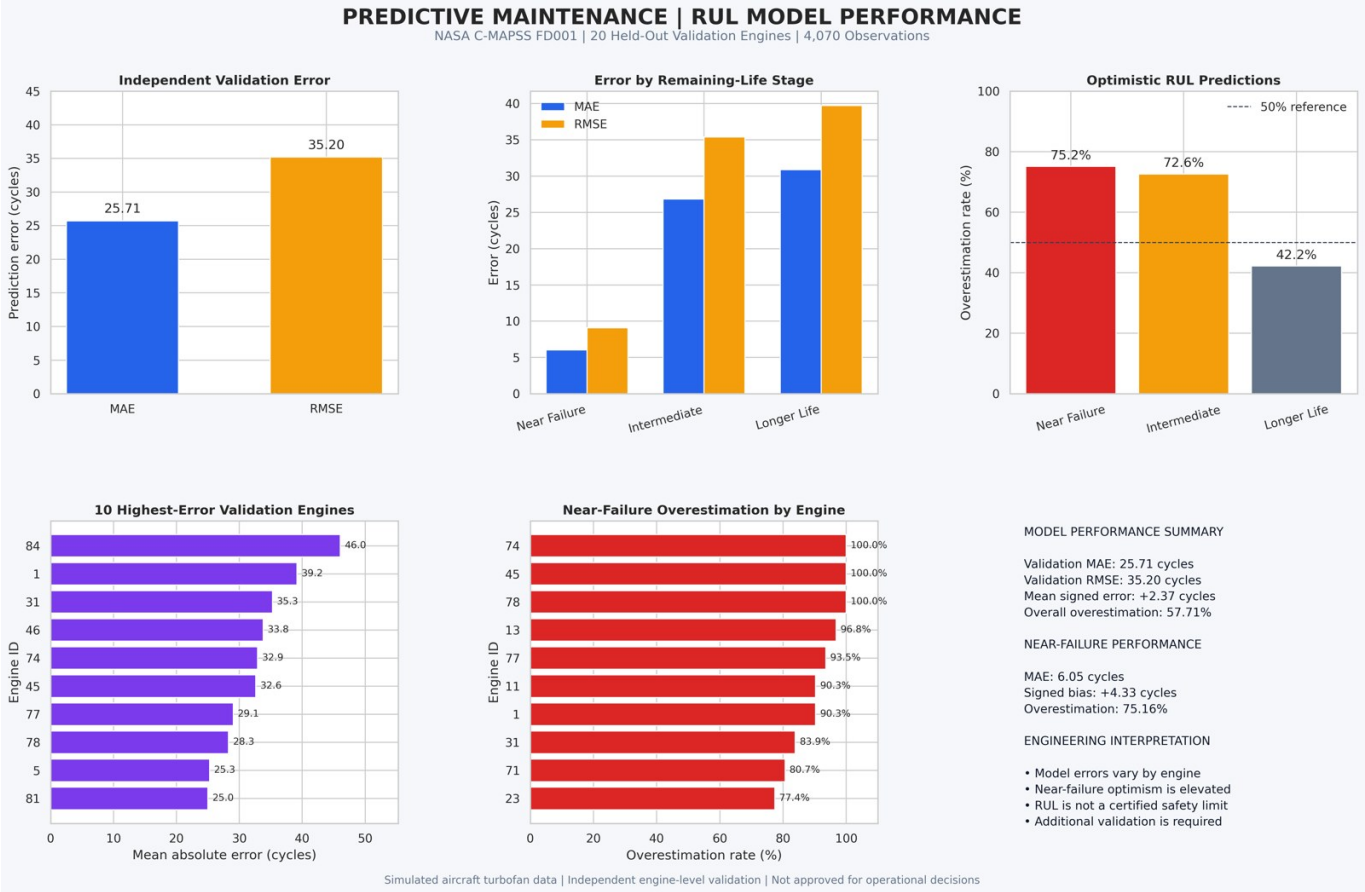
Engine Operating Life



Sensor Degradation



Predictive Maintenance



Twin-Engine Comparison

